

7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal

Signals Using Periodogram Method

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
fs = 1000;
```

```
t = (0:999)/fs;
```

```
s = sin(2*pi*100*t) + 0.5*sin(2*pi*200*t);
```

```
n = 0.3*randn(size(t));
```

```
x = s + n;
```

```
figure;
```

```
plot(t, x);
```

```
grid on;
```

```
title('Signal with Noise');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
figure;
```

```
periodogram(x, [], [], fs);
```

```
title('PSD using Periodogram');  
  
figure;  
X = abs(fft(x));  
f = (0:length(X)-1)*fs/length(X);
```

```
plot(f(1:500), X(1:500));  
grid on;  
title('Magnitude Spectrum');  
xlabel('Frequency (Hz)');  
ylabel('Magnitude');
```

```
figure;  
bar([mean(s.^2) var(n)]);  
set(gca, 'XTickLabel', {'Signal', 'Noise'});  
title('Power Comparison');  
grid on;
```

Output:



